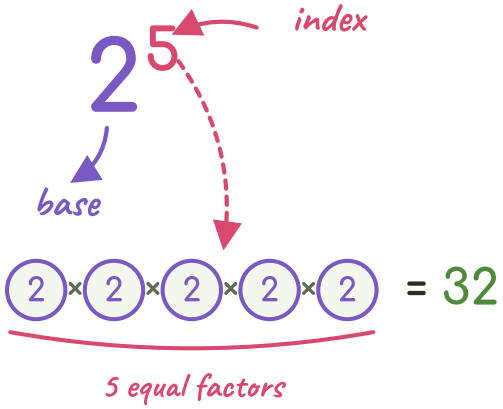
Powers on one page

Hi! Here is everything important about powers: what they are and three rules that make hard sums easy. Follow the arrows.



What a power is 1

A power is a short way to write repeated multiplication: $a^n = a \times a \times \ldots \times a$ (n times).



$a^1 = a$: one factor is just the number. Say it: "two to the power of five".

Minus signs and brackets 2

A negative number to an **even** power turns positive; to an **odd** power it stays negative.

$(-2)^2 = 4$ — *even → plus*

$(-2)^3 = -8$ — *odd → minus*

$-2^2 = -4$ — *minus outside!*

-2^2 means $-(2^2)$: the power only belongs to the 2. To include the minus, use brackets.



Brackets are like a hug: everything inside gets the power.

Key rule: multiplying powers 3

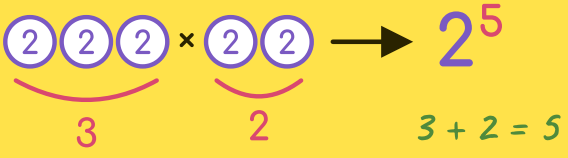
Same base? Add the indices: $a^m \times a^n = a^{m+n}$.

This only works for the same base: you can't do it with $2^3 \times 5^2$.



Same base? Put them all in one group and count!

$2^3 \times 2^2 =$

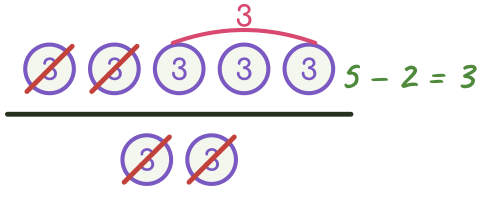


check: 8 × 4 = 32 = 2^5 ✓

Dividing powers 4

Same base? **Subtract the indices**: $a^m \div a^n = a^{m-n}$ $(a \neq 0)$.

$3^5 \div 3^2 = 3^3 = 27$



pairs on top and bottom cancel

Check: 243 ÷ 9 = 27.

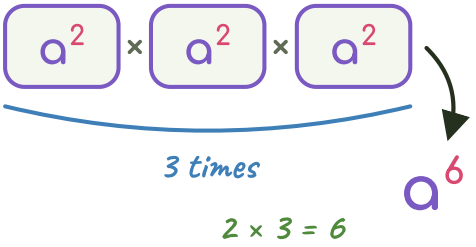


Cancel the pairs and there's less work to do!

A power of a power 5

Raise a power to a power? **Multiply the indices**: $(a^m)^n = a^{mn}$.

$(a^2)^3 =$



Don't mix them up: $a^2 \times a^3 = a^5$ (add), but $(a^2)^3 = a^6$ (multiply).

6

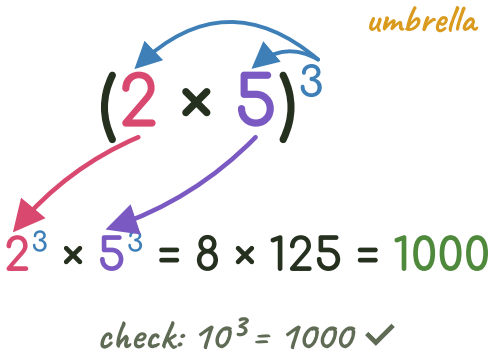
A power of a product

The index outside the brackets goes to **every** factor: $(ab)^n = a^n b^n$.

It works backwards too: $25^2 \times 4^2 = (25 \times 4)^2 = 100^2 = 10{,}000$. Same index? Put them under one bracket.



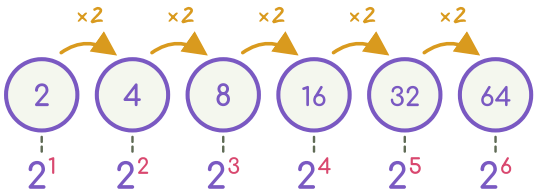
The index outside is an umbrella over everyone.



Change to the same base

7

Different numbers are often powers of the same base. Rewrite them, and the rules work again.



each step: multiply by 2 once more

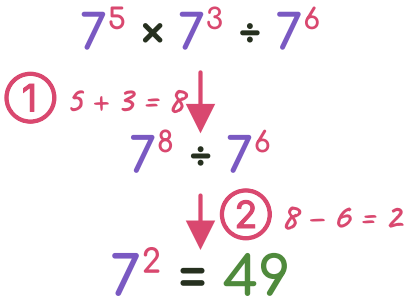
$8^2 \times 4 = (2^3)^2 \times 2^2 = 2^6 \times 2^2 = 2^8 = 256$

Handy to know: $9 = 3^2$, $27 = 3^3$, $49 = 7^2$, $100 = 10^2$, $1000 = 10^3$.

Simplify first, then work it out

8

Don't multiply out huge numbers. Combine the indices first, then work out a small power.





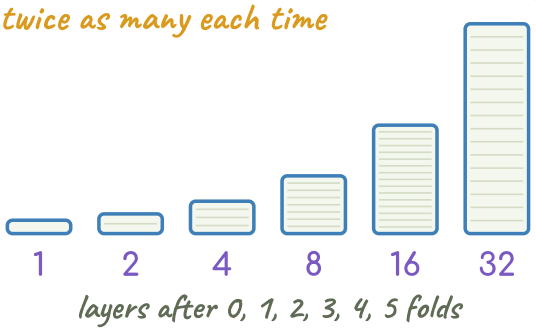
7^8 is over 5 million. Why bother, when the answer is 49?

Powers in real life

9

Fold a sheet of paper in half and there are twice as many layers. Fold again: twice as many again.

Five folds make $2^5 = 32$ layers. Ten make $2^{10} = 1024$. Rumours spread just as fast when everyone tells two friends.





Check yourself: same base? Multiplying: add the indices. Dividing: subtract them. Power of a power: multiply them.

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Built on his slides and lessons.